Multiples of 10 involving negative exponents

Task A: Introducing negative exponents

1) Fill in the gaps:

	Decimal	Fraction	Exponential Form
a	0.09	$\frac{9}{100}$	9×10^{-2}
b	0.00004	$\frac{4}{100\,000}$	4×10^{-5}
c	0.3	$\frac{3}{10}$	3×10^{-1}
d	0.2	$\frac{2}{10}$	2×10^{-1}
e	0.0007	$\frac{7}{10\,000}$	7×10^{-4}
f	0.000008	$\frac{8}{1\,000\,000}$	8×10^{-6}
g	0.007	$\frac{7}{1000}$	7×10^{-3}
h	0.0005	$\frac{5}{10\,000}$	5×10^{-4}

2) What mistake has Aisha made?

Write 0.0003 in expanded exponential form.

Aisha writes:

$0.0003 = 3 \times 10^{-3}$ because there are 3 zeros after the decimal point.

Aisha

The 3 is in the ten thousandths column.

10^{-3} is one thousandth not one ten thousandth.

Therefore, the answer should be 3×10^{-4}

3) Write each of the following in the alternative format:

	Decimal	Exponential form
a	0.34	$3 \times 10^{-1} + 4 \times 10^{-2}$
b	0.52	$5 \times 10^{-1} + 2 \times 10^{-2}$
c	0.0473	$4 \times 10^{-2} + 7 \times 10^{-3} + 3 \times 10^{-4}$
d	0.049	$4 \times 10^{-2} + 9 \times 10^{-3}$
e	0.000603	$6 \times 10^{-4} + 3 \times 10^{-6}$
f	0.200303	$2 \times 10^{-1} + 3 \times 10^{-4} + 3 \times 10^{-6}$
g	17.02	$1 \times 10^{1} + 7 \times 10^{0} + 2 \times 10^{-2}$
h	200.3058	$2 \times 10^{2} + 3 \times 10^{-1} + 5 \times 10^{-3} + 8 \times 10^{-4}$

Task B: Writing integers in a range of forms

1) Identify the incorrect alternative in each row. Record the letter and rearrange them to make a word. *EXPONENT*

Number	Alternative 1		Alternative 2		Alternative 3		Alternative 4	
3000	$\boxed{3 \times 10^4}$	N	$3 \times 10^2 \times 10$	L	$\frac{30}{10} \times 10^3$	I	$\frac{300}{10} \times 10^2$	Y
74 000	7.4×10^4	A	74×10^3	P	$\boxed{7.4 \times 10^3}$	X	$\frac{740\ 000}{10}$	C
120 000	$12 \times 10^2 \times 10^2$	R	1.2×10^5	H	$\boxed{1.2 \times 10^4}$	E	$1.2 \times 10^6 \times 10^{-1}$	B
450	4500×0.1	S	$\frac{450}{10^3} \times 10^3$	E	4.5×10^2	F	$\boxed{4.5 \times 10^{-2}}$	O
8 000 000	$8 \times 10^3 \times 10^3$	T	$\boxed{\frac{8}{10^3} \times 10^8}$	P	8×10^6	V	$\frac{8000}{10^2} \times 10^5$	D
628 000	$6\ 280\ 000 \times 10^{-1}$	O	6.28×10^5	D	$\boxed{6.28 \times 10^3}$	N	628×10^3	P
32 490	$\boxed{32.49 \times 10^2}$	E	324.9×10^2	M	$\frac{3249}{10^4} \times 10^5$	A	$324\ 900 \times 10^{-1}$	H
12 400 000	$12.4 \times 10^{-3} \times 10^9$	N	$\frac{124}{10^2} \times 10^7$	L	1.24×10^7	O	$\boxed{124 \times 10^{-2} \times 10^6}$	T